> Before starting, make sure you **read all the questions** and understand what is asked of you. Write your answers to the questions in a separate sheet of paper and make sure you **justify all of your answers**. Multiple-choice questions have a **unique correct answer**.

1. (5 points) Which one of the following is an advantage of hand-built scanners over automata-based scanners?
   - A. They can always be automatically constructed.
   - B. They can go beyond regular languages.
   - C. They implicitly factor out common prefixes.
   - D. They require less coding effort.

2. (5 points) Which one of the following languages is not a regular language?
   - A. The set of all words on $\Sigma = \{(,)\}$ such that every) matches a previous (and every (is eventually matches to a).
   - B. The set of all prefixes of the string "Compilers is the best course in our program".
   - C. The set of all first names of people born in 2022.
   - D. The set of all Dutch words.

3. (5 points) Argue that the following language over the alphabet $\Sigma = \{0, 1, \#\}$ is context free:

   $$L_{\text{pal}\#} = \{w\#w^R \mid w \in \Sigma^*\}$$

   where w^R denotes the word spelled in reverse order.

4. (10 points) Construct the canonical finite-state machine for the following grammar with terminal set $T = \{a, b, c, d, \$\}$.

(1)	S	$\rightarrow$	$A\$$
(2)	A	$\rightarrow$	aCD
(3)		$\rightarrow$	ab
(4)	C	$\rightarrow$	c
(5)	D	$\rightarrow$	d

5. (10 points) Is the **language** from the previous exercise LR(1)?

6. (10 points) Give an arithmetic-expression tree such that the minimal number of registers required to evaluate it is exactly 7.

7. (5 points) Label the following instructions with the set L of live variables and a function U from variables to their next usage.

```
y := 1 + 2
x := 5
w := x + y
z := (x − w) ∗ y
```

8. (5 points) Let $T = \{\texttt{cst}, \texttt{id}, +, *\}$ be a set of terminals. Is the following **grammar** LALR(1)?

(1)	Exp	$\rightarrow$	Exp + Exp
(2)		$\rightarrow$	Exp ∗ Exp
(3)		$\rightarrow$	(Exp)
(4)		$\rightarrow$	`id`
(5)		$\rightarrow$	`cst`

9. (15 points) Consider the same set of terminals as in the previous question. Give an LALR(1) grammar whose language is the same as the one given below. (Remember to argue that the languages of the two grammars are the same!)

(1)	Exp	$\to$	Exp + Prod
(2)		$\to$	Prod
(3)		$\to$	(Exp)
(4)	Prod	$\to$	Prod $*$ Atom
(5)		$\to$	Atom
(6)		$\to$	(Prod)
(7)	Atom	$\to$	`cst`
(8)		$\to$	`id`
(9)		$\to$	(Exp)

10. (10 points) Argue that the converse of the following statement studied during the course is not true.

Theorem 1 (Sentential-form prefixes on the stack)**.** *Let $(q, w, \gamma Z_0)$ be a configuration reached along an accepting run of the classical bottom-up parser with q not of the form $(A, \dots)$. Then:*

$$S \Rightarrow^* \gamma^R \cdot w$$

along a rightmost derivation.

In other words: Give an example of a context-free grammar G and a rightmost derivation of it that reaches a sentential form $\gamma^R w$ such that $(q, w, \gamma Z_0)$ is **not** a configuration reached along an accepting run of the classical bottom-up parser constructed for G — i.e. the one that only has Shift/Reduce transitions.

11. (10 points) Give an example of an LR(0) **language** that is not LL(1).

12. (10 points) Give an example of a context-free **grammar** that is not LR(1).